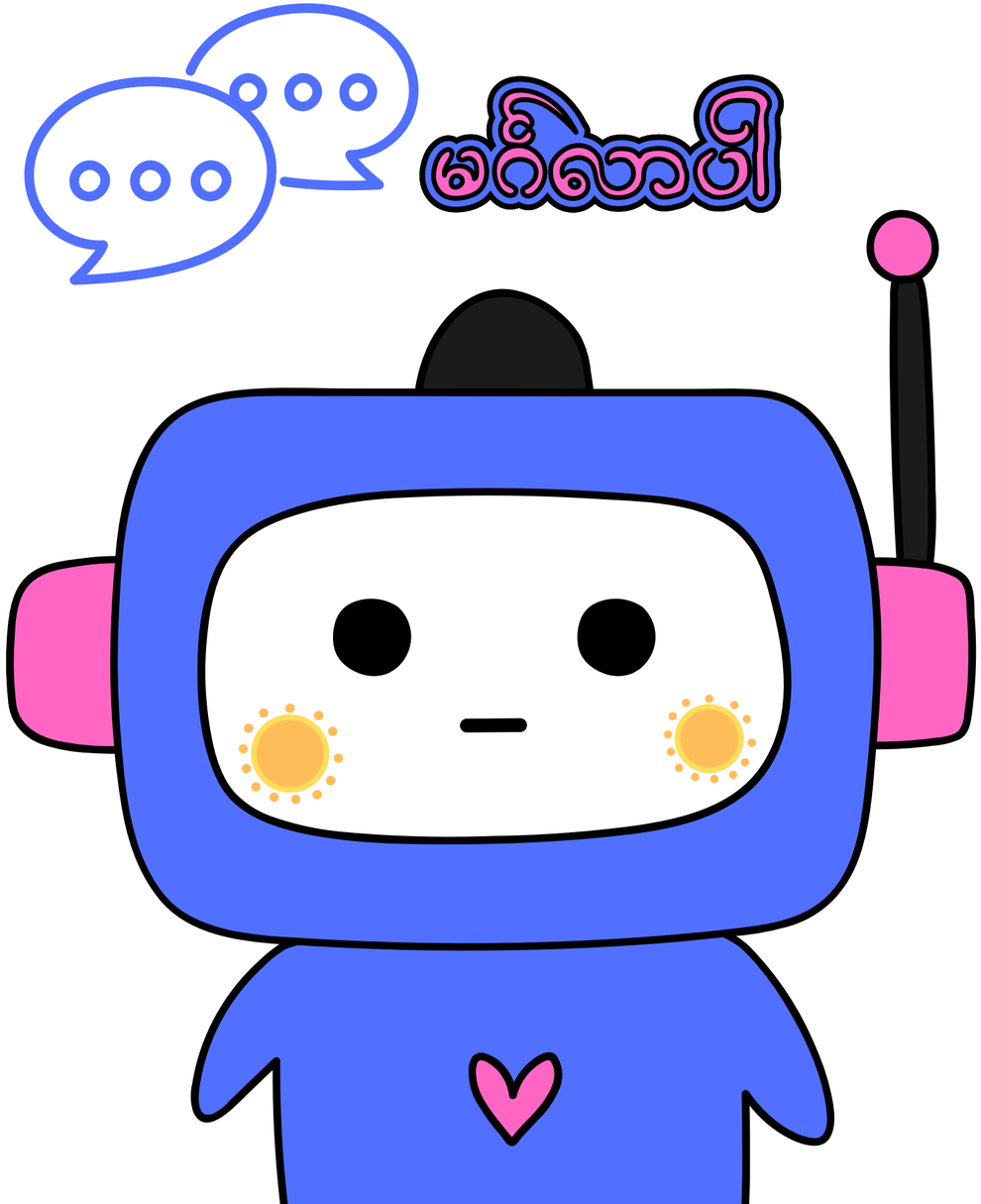
Burmese Elyza

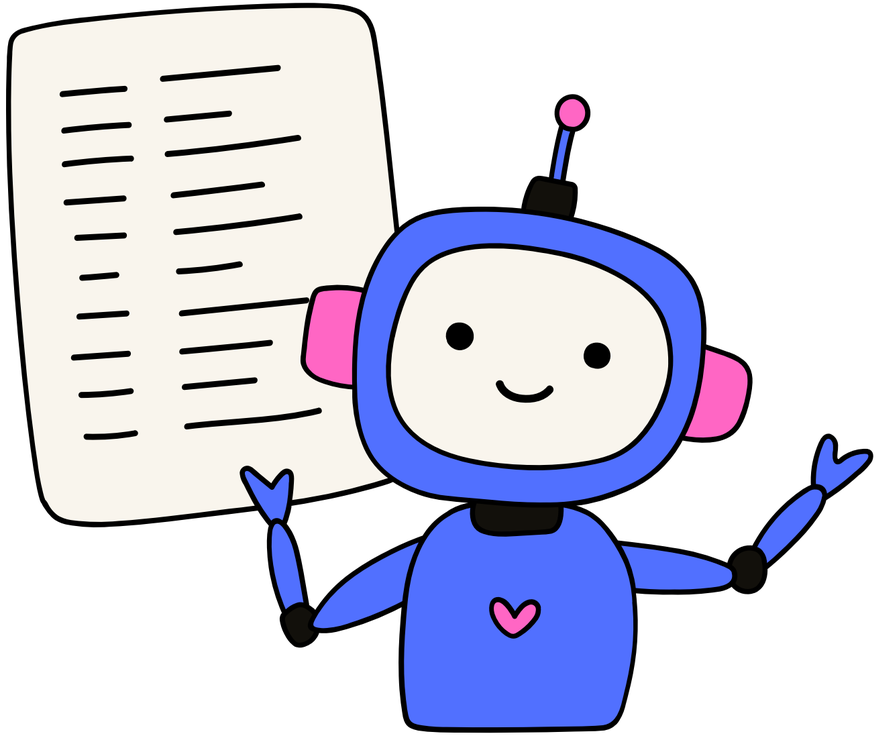
Group 6

- Htet Kaung San
- Aung Nanda Oo
- Thet Hnin Phyu
- Min Pyae Hein
- Khin Kaung San Thu Tun
- Hsu Yee Win
- Eaint Lay Hmone
- Htoo Min Htet
- Akeri Myint Zaw
- Sai Zay Linn Htet



Agenda

- Data Collection
- Data Exploratory Analysis
- Data Preprocessing
- Model Training
- Evaluating Model Performance
- Pattern Matching



Data Collection

6 emotion classes

Sadness · Joy · Love · Anger · Fear · Surprise

Manual Data

- 1203 rows for **total**
- 200 for **each class**

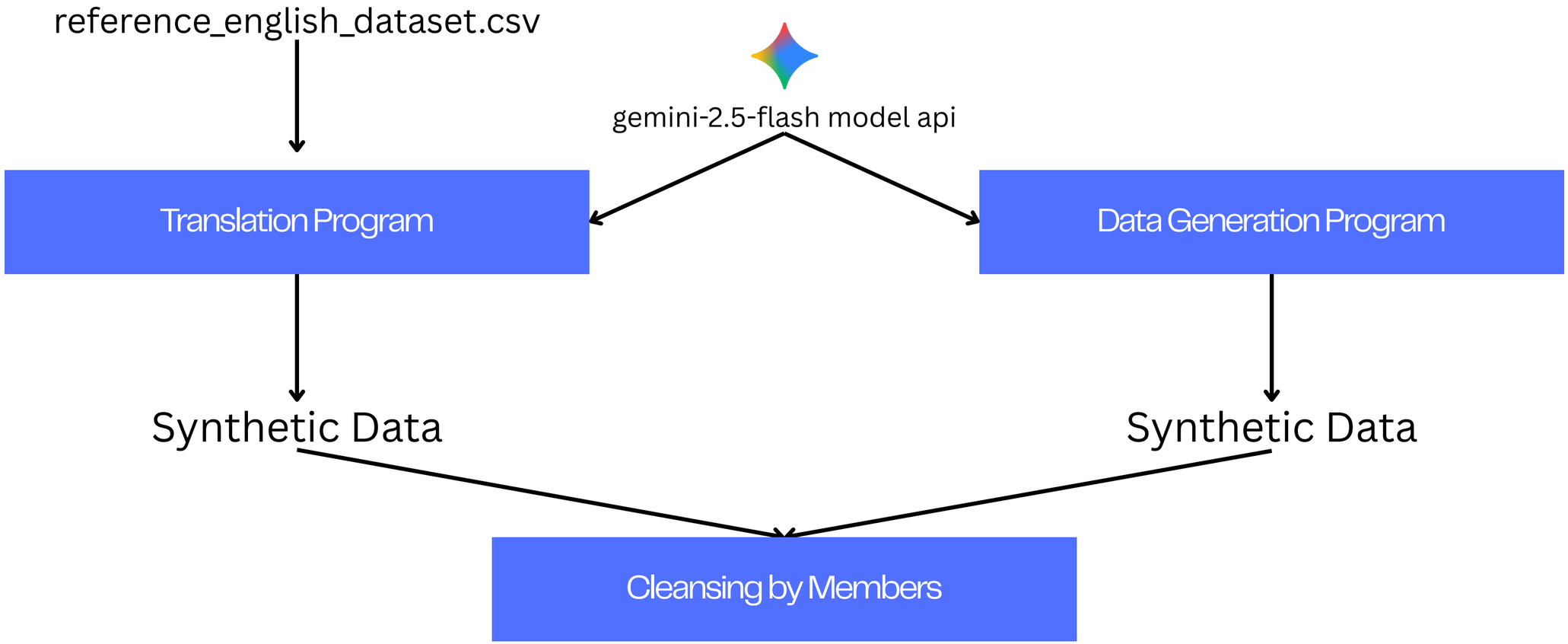
Synthetic Data

- Generated Data
- Translation Data
  - gemini-2.5-flash-lite

5220 rows for **total**

Two Approaches

to get Synthetic data



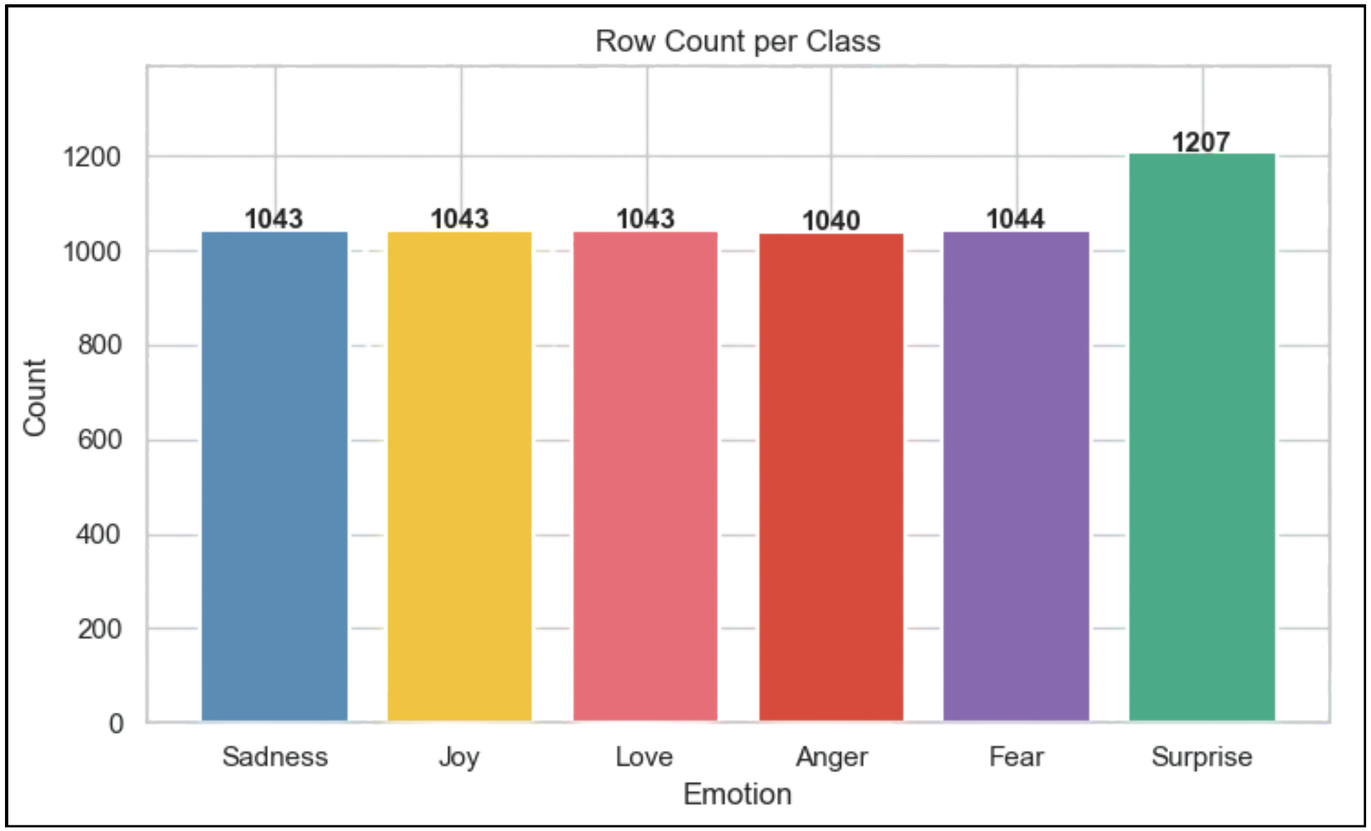
Dataset Format

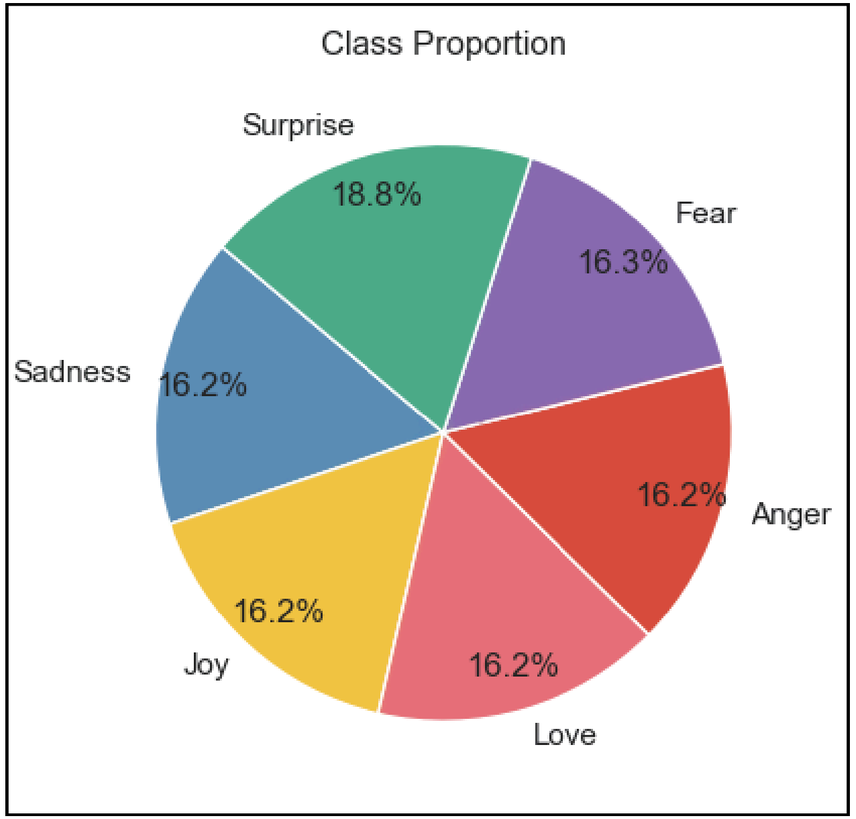
Text	Label
ဘာအကြောင်းပြချက်မှ မရှိဘဲ ဝမ်းနည်းနေတယ်။	0
ခရီးလမ်းတစ်လျှောက် မြင်ရတဲ့ ရှုခင်းတွေက ဓာတ်ပုံထဲမှာထက် အပြင်မှာ ပိုလှနေလို့ အရမ်းပျော်မိတယ်	1
လူတစ်ယောက်ရဲ့ဘဝမှာအများကြီးမလိုအပ်ဘူး တကယ်လိုအပ်တာဟာသူ့ကိုချစ်ပေးမဲ့သူနဲ့ သူဘက်ကချစ်ရတဲ့သူပဲလိုတာ။	2
အခုလို ပြန်ရေးတာ မြင်ရင်တောင် ဟိုက တိတ်တိတ်လေး ဒေါသထွက်ရင်ထွက်နေမှာ။	3
YBSပေါ်စပါးကြီးမြွေတွေနဲ့ စီးနင်းနေတာရှိတာကြောင့် ခရီးသည်တွေ ကြောက်လန့်ဖယ်ပေးနေရ	4
ဆရာက ရုတ်တရက်ကြီး စာမေးပွဲရှိတယ် ပြောလိုက်တော့ လန့်သွားတာပဲ	5

{0:"Sadness",
1:"Joy",
2:"Love",
3:"Anger",
4:"Fear",
5:"Surprise"}

EDA

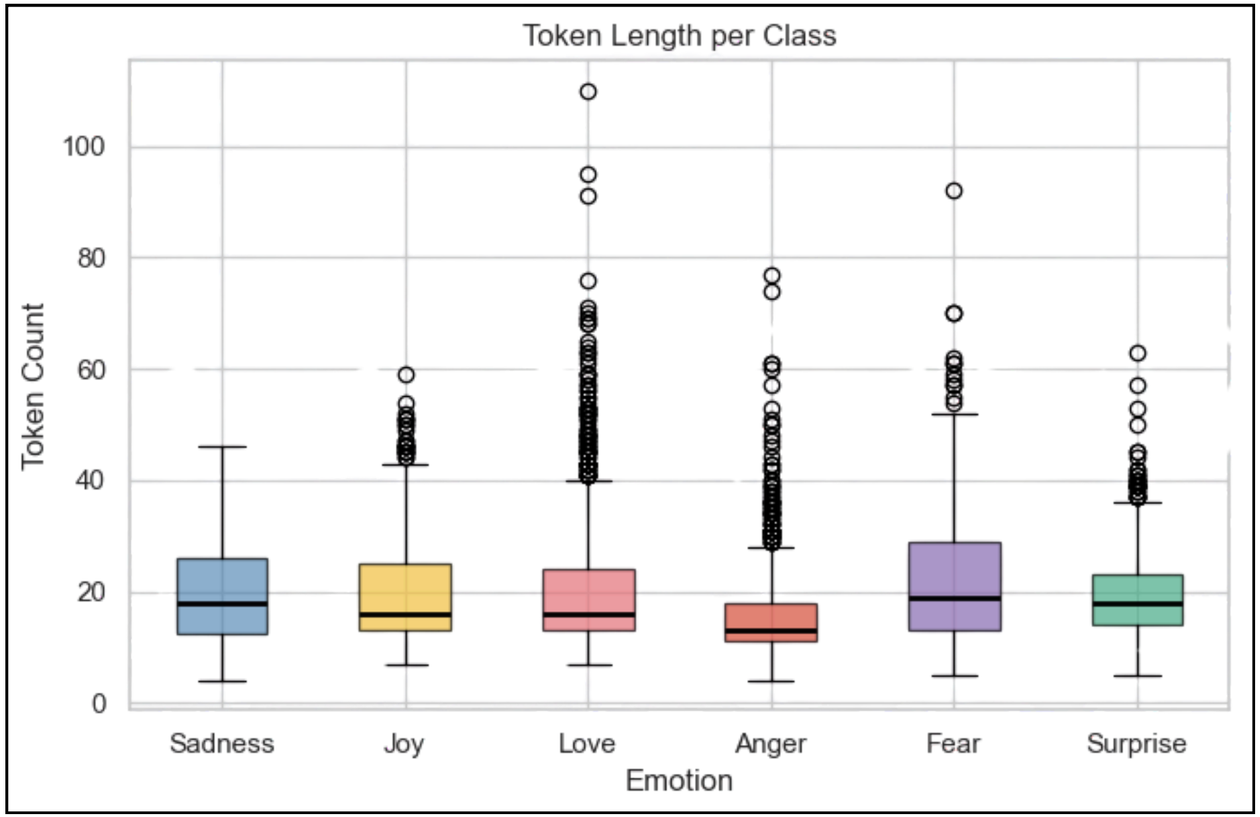
Class Distribution

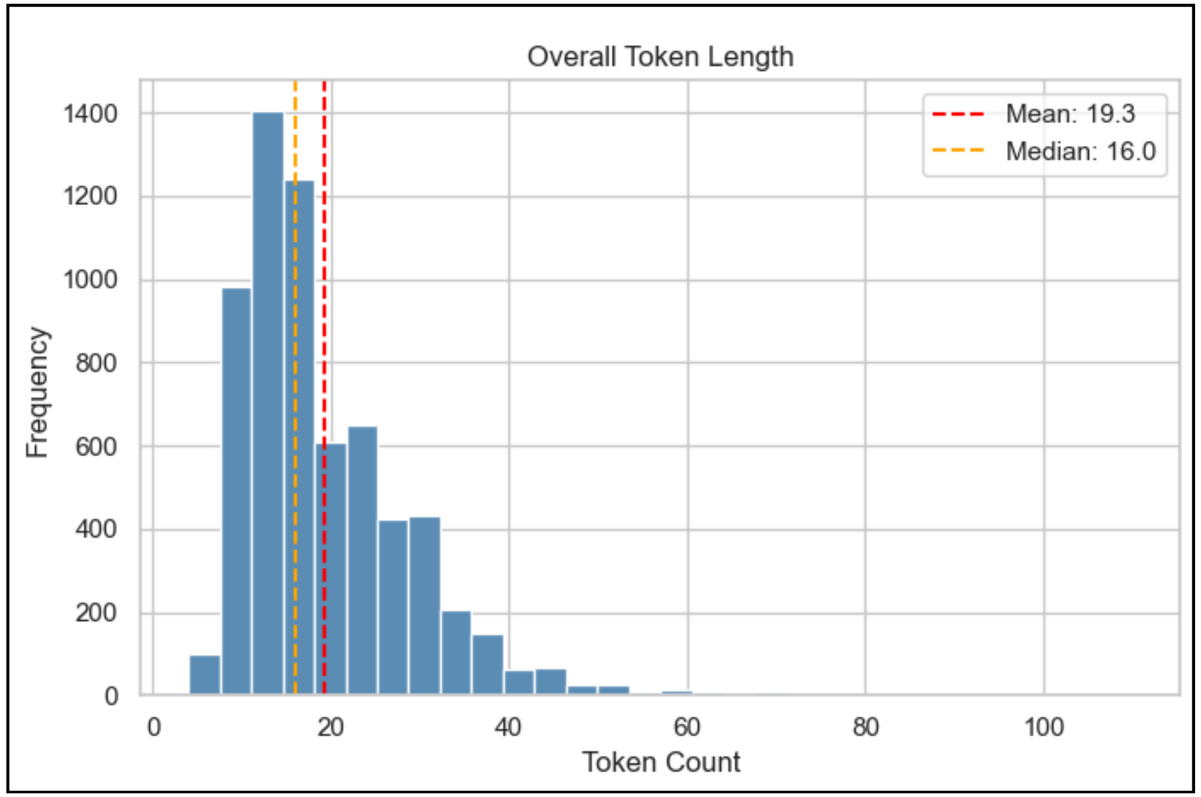




EDA

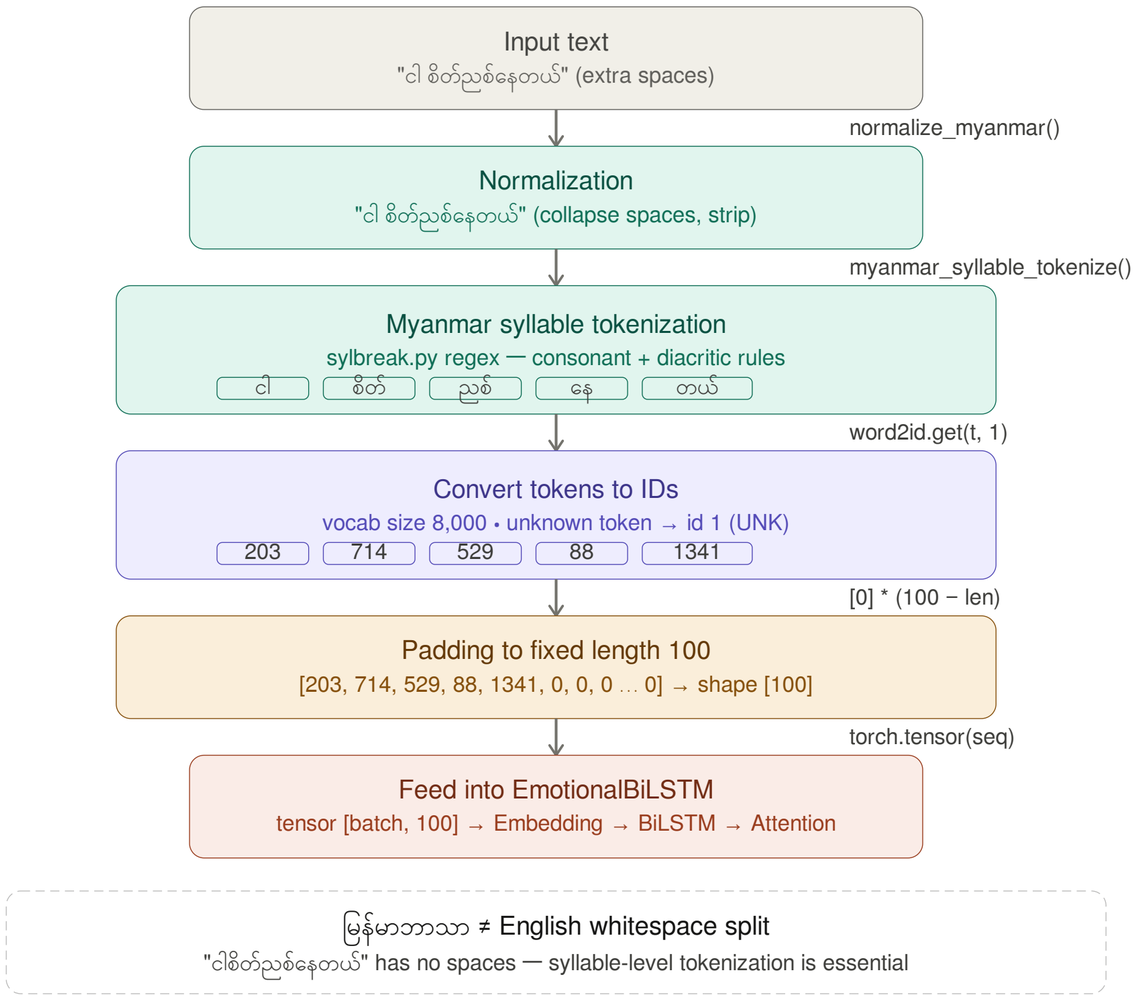
Text Length Analysis





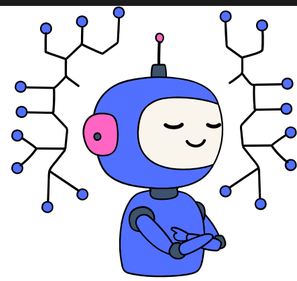
Data Preprocessing

- Normalize input text (remove extra spaces)
- Tokenize using Myanmar syllable segmentation (sylbreak)
- Convert tokens into numerical IDs (word2id)
- Pad sequences to fixed length (100)



Reference - https://github.com/ye-kyaw-thu/sylbreak

image generated with claude



Model

BiLSTM with Attention

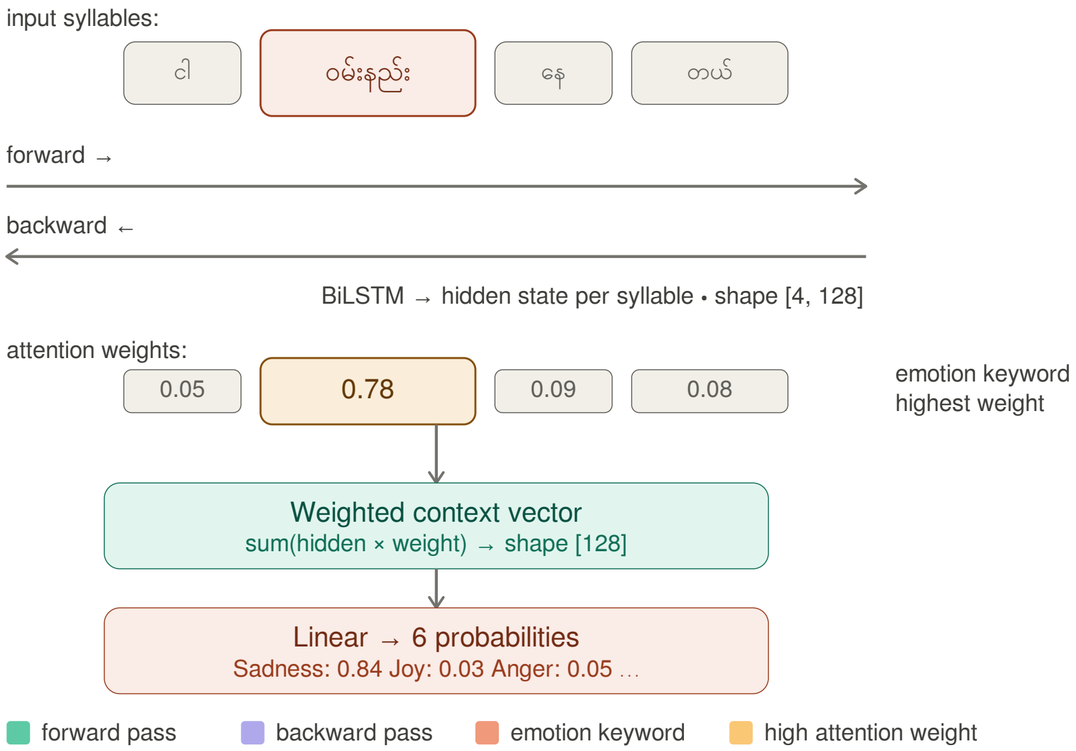


image generated with claude

Model

Training Parameter

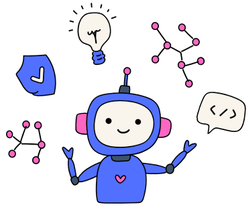
Key Parameters

- Epochs: 20
- Batch Size: 32
- Learning Rate: 0.001
- Validation Split: 20%
- Max Sequence Length: 100

Optimizer : Adam

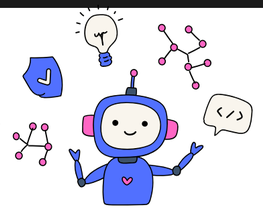
Loss Function : CrossEntropyLoss

Evaluating Model Performance



Embedding	Model	Dataset	Training Accuracy	Validation Accuracy	Note
FastText	BiLSTM	Manual	-	-	-
Syllbreak + word2id	BiLSTM	Manual	100%	81.67%	Overfit
FastText	BiLSTM	Manual + Synthetic	-	-	-
Syllbreak + word2id	BiLSTM	Manual + Synthetic	99.34%	90.97%	-

Testing Accuracy



Generated with Data Generation Program
(Gemini-flash-2.5 model api)

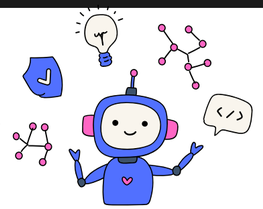
```
============================================================
Classification Report (Test Set — 600 samples)
============================================================
              precision    recall  f1-score   support

     Sadness     0.7101    0.4900    0.5799       100
         Joy     0.8495    0.7900    0.8187       100
        Love     0.5053    0.9500    0.6597       100
       Anger     0.8281    0.5300    0.6463       100
        Fear     0.8990    0.8900    0.8945       100
    Surprise     0.8621    0.7500    0.8021       100

    accuracy                         0.7333       600
   macro avg     0.7757    0.7333    0.7335       600
weighted avg     0.7757    0.7333    0.7335       600

Overall Test Accuracy: 73.33%
```

Testing Accuracy



```
  AGGREGATED RESULTS  (5-Fold Cross-Validation)
==============================================================

Per-fold accuracies : ['89.95%', '91.36%', '89.72%', '89.49%', '90.89%']
Mean Accuracy       : 90.28%
Std  Accuracy       : 0.72%
Best fold Accuracy  : 91.36%

Combined Classification Report (all held-out predictions):
              precision    recall  f1-score   support

     Sadness     0.8716    0.8591    0.8653      1043
         Joy     0.9012    0.9099    0.9055      1043
        Love     0.9276    0.9463    0.9369      1043
       Anger     0.8837    0.8692    0.8764      1040
        Fear     0.9011    0.8898    0.8954      1044
    Surprise     0.9263    0.9370    0.9316      1207

    accuracy                         0.9028      6420
   macro avg     0.9019    0.9019    0.9019      6420
weighted avg     0.9025    0.9028    0.9026      6420
```

Pattern Matching

Rule-Based Approach

- Uses regular expressions (regex) to match user input
- Each rule has a Pattern
- Response templates
- Priority rank (0 – 10)

Multi-Level Design

- Level 1–2 → Safety, crisis
- Level 3–4 → Emoji, symbols
- Level 5–6 → Health, positive emotions
- Level 7–8 → Mental states, advice
- Level 9–10 → Short reply, fallback

Ranked Rule System

- Rules sorted by priority rank
- Higher rank → matched first
- Safety handling always first
- Correct response selection

Example Rule

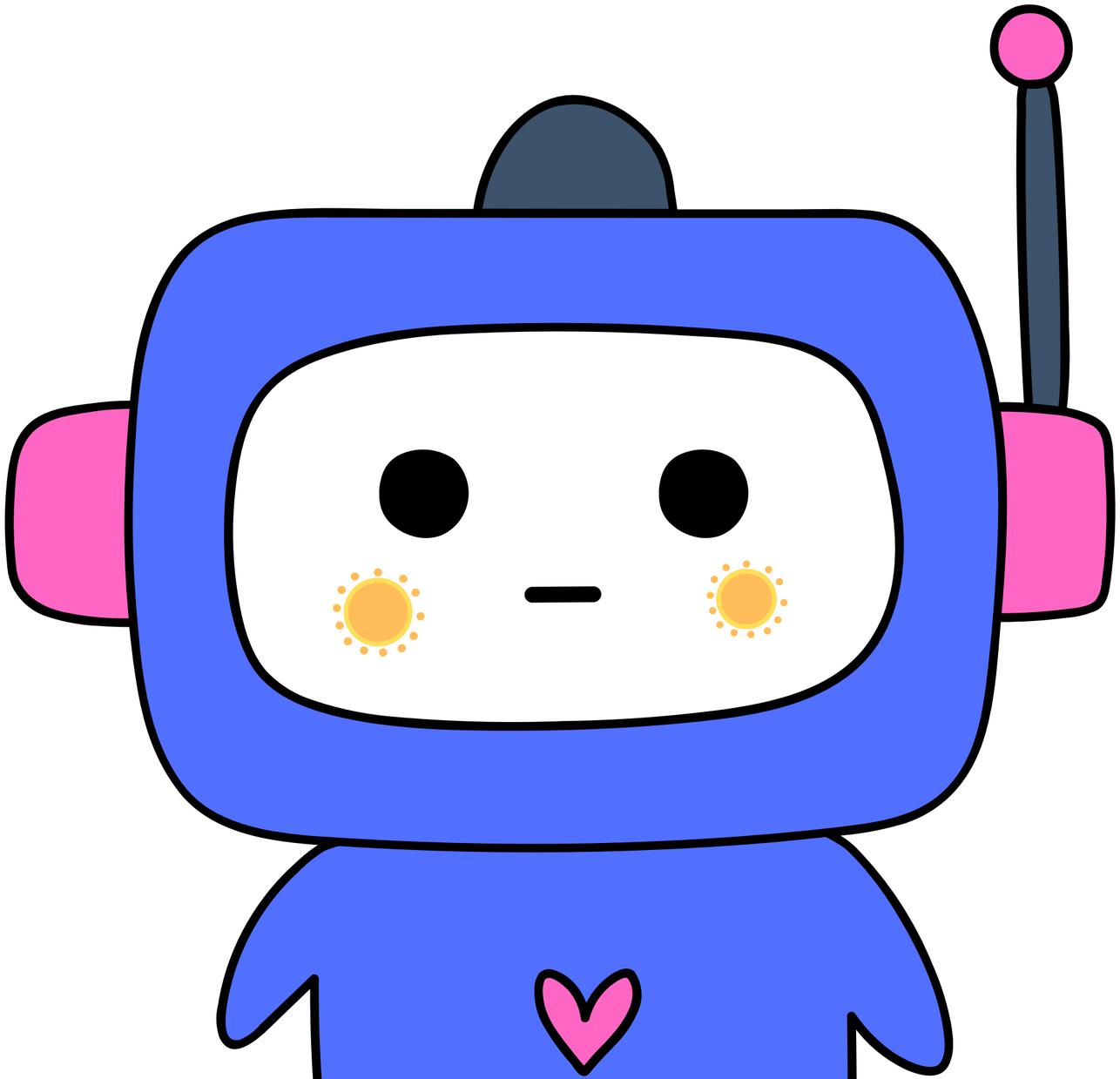
Input

ငါသေချင်တယ် 😟

Matched:

Level 1 Safety — rank 10

Response

အခုလို ခံစားနေရတာကို ကြားရတာ စိုးရိမ်မိပါတယ်ဗျာ။ စိတ်ကို ဖြေလျှော့ပြီး အကူအညီတောင်းကြည့်ဖို့ တိုက်တွန်းချင်ပါတယ်။ အချိန်မရွေး ရင်ဖွင့်လို့ ရပါတယ်နော်။

THANK YOU